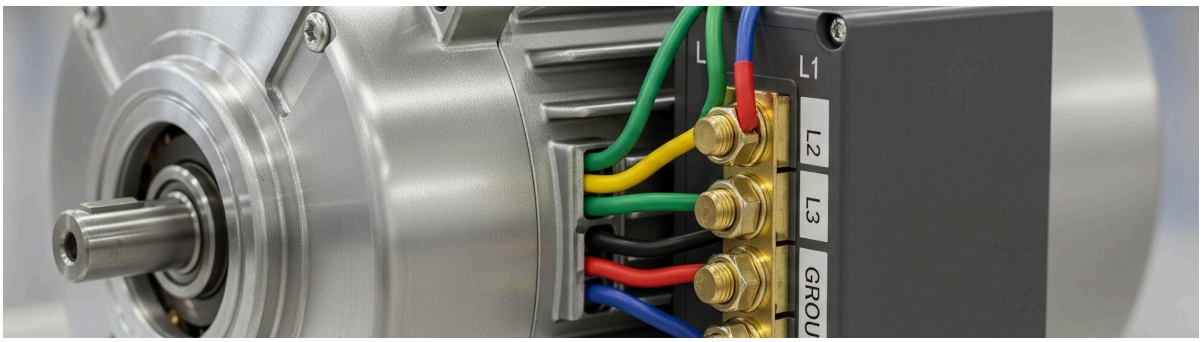

Electrical Machine Data Sheet

Machine Name/Type : Capacitor Start Capacitor Run Single Phase Induction Motor

Name plate Details

The rating section details the machine's operational capacity and design limits.

PARAMETER	VALUE	UNIT
Power output	2	HP
Voltage	230	V
Rated Current	9.9	A
Speed	1440	RPM
Frequency	50	Hz
Start Capacitor	200, 275	µF, V
Run Capacitor	15, 440	µF, V

Precautions

I. Ensure the supply voltage is **230 V AC single phase** before starting.

II. Do not operate the motor if **any capacitor is disconnected or damaged**.

III. Ensure correct connection of start and run capacitors as per circuit diagram.

IV. Avoid frequent switching ON and OFF.

V. Do not overload the motor beyond rated capacity.

VI. Ensure proper earthing of the motor body.

VII. Maintain adequate ventilation for cooling.

VIII. Check terminal connections for tightness before operation.

IX. Keep hands and tools away from rotating parts.

X. Switch OFF immediately in case of overheating, abnormal noise, or vibration.

Working Principle of Capacitor Start Capacitor Run Motor

When a single-phase AC supply is applied, the main winding produces a **pulsating magnetic field**, which is not sufficient to start the motor.

An **auxiliary winding** is provided with a **starting capacitor** and a **running capacitor** to create a phase difference between currents.
This phase difference produces a **rotating magnetic field**, generating high starting torque.

After the motor reaches near rated speed, the starting capacitor is disconnected using a centrifugal switch, while the running capacitor remains in the circuit to improve power factor and efficiency.

Working Characteristics

I. High starting torque compared to other single-phase motors.

II. Smooth and quiet operation.

III. Better efficiency and power factor.

IV. Nearly constant speed under load.

V. Suitable for continuous-duty applications.

VI. Reduced vibration due to improved phase balance.

Applications

1. Air compressors
2. Refrigeration and air-conditioning units
3. Water pumps
4. Washing machines
5. Conveyors
6. Machine tools
7. Oil burners
8. Industrial fans and blowers
9. Small hoists
10. Agricultural equipment